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Electric guitar body-structure and electric guitar

Abstract

A body structure of an electric guitar includes a body including a first chamber and a second chamber formed spaced apart from each other, and a slit that connects the first chamber and the second chamber.

Inventors:	Matsuda; Hideto (Hamamatsu, JP), Ishizaka; Kenta (Hamamatsu, JP)
Applicant:	YAMAHA CORPORATION (Hamamatsu, JP)
Family ID:	1000008818373
Assignee:	YAMAHA CORPORATION (Hamamatsu, JP)
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Primary Examiner: Lockett; Kimberly R

Attorney, Agent or Firm: ROSSI, KIMMS & McDOWELL LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of International Application No. PCT/JP2020/045271, filed Dec. 4, 2020, which claims priority to Japanese Patent Application No. 2019-220464, filed Dec. 5, 2019. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

- (1) The present disclosure relates to an electric guitar body-structure and an electric guitar.
- (2) Priority is claimed on Japanese Patent Application No. 2019-220464, filed Dec. 5, 2019, the content of which is incorporated herein by reference.

Description of Related Art

- (3) Japanese Unexamined Patent Application, First Publication No. 2001-154662 (hereinafter referred to as Patent Document 1) discloses a technique capable of producing a beautiful and reverberant sound by providing a plurality of grooves on the inner side of the body plate of the hollow body of a stringed instrument such as an acoustic guitar or a violin. A beautiful and reverberant sound is obtained by appropriately controlling the acoustic phenomenon of the body.

SUMMARY OF THE INVENTION

- (4) In some cases, a plurality of chambers (cavities) are provided in the interior of bodies of electric guitars and the like for weight reduction. However, if there are a plurality of chambers having substantially the same volume in this type of body, the resonance frequencies of the plurality of chambers become substantially the same, giving rise to the necessity to control the acoustic phenomenon of the body. It is difficult to control the acoustic phenomenon of the body even if a plurality of grooves as in Patent Document 1 are provided on the inner surface of the chambers of the body of the electric guitar.
 - (5) The present disclosure has been made in view of the above circumstances, and has as its object to provide an electric guitar body-structure capable of controlling the acoustic phenomenon of a body having a plurality of chambers, and an electric guitar including the same.
 - (6) A first aspect of the present disclosure is an body structure of an electric guitar, including: a body including a first chamber and a second chamber formed spaced apart from each other, and a slit that connects the first chamber and the second chamber.
 - (7) A second aspect of the present disclosure is an electric guitar including the body structure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plan view of an electric guitar according to an embodiment of the present disclosure as viewed from the front side of the body.
- (2) FIG. 2 is a plan view of a back member constituting the body of the electric guitar of FIG. 1 as viewed from the front side thereof.
- (3) FIG. 3 is an enlarged perspective view showing two chambers and slits in the back member of FIG. 2.
- (4) FIG. 4 is an enlarged plan view showing main parts of an electric guitar body according to another embodiment of the present disclosure.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

- (5) Hereinbelow, embodiments of the present disclosure will be described with reference to FIGS. 1 to 3.
- (6) As shown in FIG. 1, an electric guitar 1 according to the present embodiment includes a body structure 2, a neck 3, and strings 4.
- (7) The neck 3 is connected to an end portion of the body structure 2 and extends in a direction away from the body structure 2 (upward direction in FIG. 1). A head 5 forming the distal end portion in the longitudinal direction of the neck 3 is provided with pegs 6 around which an end portion of each of the strings 4 is wound. The strings 4 are stretched along the longitudinal direction of the neck 3.
- (8) The body structure 2 includes a body 20. In this embodiment, the body 20 constitutes the entire

body structure **2**. A bridge **7**, an electromagnetic pickup **8**, controllers, and the like are attached to the body **20**. The bridge **7**, the electromagnetic pickup **8**, and the controllers are exposed on a front side surface **20a** (hereinafter referred to as the front surface **20a**) of the body **20** facing the thickness direction of the body **20** (direction orthogonal to the paper surface in FIG. **1**).

(9) One end of each string **4** is fastened to the bridge **7**. The electromagnetic pickup **8** is located between the neck **3** and the bridge **7** in the longitudinal direction of the neck **3**. A plurality of the electromagnetic pickups **8** (two in the illustrated example) are arranged in the longitudinal direction of the neck **3**. The controllers adjust the volume, tone, and the like of the acoustic signal output from the electromagnetic pickups **8**. The controllers include two volume switches **9**, a pickup selector **10** for switching the electromagnetic pickups **8** to be activated, and the like.

(10) The body **20** of the present embodiment has a top member **21** having a small thickness dimension and a back member **22** (see FIGS. **2** and **3**) having a thickness dimension larger than that of the top member **21**. The top member **21** and the back member **22** overlap with each other in the thickness direction of the body **20** to form the body **20**. The front surface **20a** of the body **20** from which the bridge **7** and the like are exposed is constituted by the top member **21**.

(11) As shown in FIGS. **2** and **3**, the body **20** has a plurality of chambers **24** (**19** in the illustrated example) and slits **25**.

(12) The plurality of chambers **24** are cavities formed for weight reduction of the body **20**. The plurality of chambers **24** are formed at intervals from each other. Specifically, the plurality of chambers **24** are arranged in a direction orthogonal to the thickness direction of the body **20**. The plurality of chambers **24** are formed in a region of the body **20** other than the region to which the neck **3**, the bridge **7**, the electromagnetic pickups **8**, the controllers (see FIG. **1**) and the like are attached, when viewed from the thickness direction of the body **20**. Although not shown in FIG. **2**, the body **20** is also formed with holes and recess portions for accommodating the bridge **7**, the electromagnetic pickups **8**, and the controllers.

(13) In the present embodiment, the plurality of chambers **24** are each formed by being recessed from a front surface **22a** of the back member **22** facing the top member **21**. The plurality of chambers **24** become cavities that each do not open to the outside of the body **20** by superimposing the top member **21** on the front surface **22a** of the back member **22**.

(14) The slit **25** connects two chambers **24** (first chamber **24A** and second chamber **24B**) adjacent to each other among the plurality of chambers **24**. The slit **25**, similarly to the chambers **24**, does not open to the outside of the body **20**.

(15) The slit **25** extends in the arrangement direction of the two chambers **24**. The direction in which the slit **25** extends may be parallel to the arrangement direction of the two chambers **24**, or may be inclined thereto.

(16) The cross-sectional area of the slit **25** orthogonal to the arrangement direction of the two chambers **24** is smaller than each cross-sectional area of the two chambers **24** orthogonal to the arrangement direction of the two chambers **24**. The cross-sectional area of each chamber **24** used for comparison with the cross-sectional area of the slit **25** may be, for example, the cross-sectional area of the chamber **24** at the maximum in the arrangement direction of the two chambers **24**.

(17) The volume of the slit **25** is sufficiently smaller than the volume of each of the two chambers **24**.

(18) The number of slits **25** connecting the two chambers **24** may be one or two or more (a plurality). When the number of slits **25** is a plurality, the total cross-sectional area of the plurality of slits **25** is smaller than each cross-sectional area of the two chambers **24**. Further, the total volume of the plurality of slits **25** is sufficiently small as compared with each volume of the two chambers **24**.

(19) Similar to the chamber **24**, the slit **25** of the present embodiment is formed by being recessed from the front surface **22a** of the back member **22**. In FIG. **3**, the depth dimension of the slit **25** is the same as the depth dimension of the chamber **24**, but for example may be smaller than the depth

dimension of the chamber **24**. Also, the slit **25** may be formed so as not to open to the front surface **22a** of the back member **22**, for example.

(20) The volumes of the two chambers **24** (first chamber **24** and second chamber **24**) connected to each other by the slit **25** may be, for example, substantially the same. The fact that the volumes of the two chambers **24** are substantially the same means that, for example, the ratio of the volume of the second chamber **24** to the first chamber **24** is 70% or more and 130% or less.

(21) By connecting the two chambers **24** to each other with the slit **25**, a new chamber **26** (hereinbelow referred to as a composite chamber **26**) including the two chambers **24** and the slit **25** is formed. The volume of the composite chamber **26** is larger than the volume of each of the two chambers **24**.

(22) The aforementioned chambers **24** and the slit **25** will be described more specifically.

(23) As shown in FIG. **2**, **19** of the chambers **24** (**24A** to **24S**) are lined up substantially along the edge of the back member **22** as seen from the front surface **22a** side. In the following description, the numbers **1**, **2**, . . . **18** and **19** are attached in an approximately clockwise order from the chamber **24A** located at the upper right of the back member **22** (body **20**) to the chamber **24S** located at the upper left.

(24) The first and second chambers **24A** and **24B** located in the upper right portion of the back member **22** (body **20**) in FIG. **2** are connected by two slits **25Aa** and **25Ab** as shown in FIGS. **2** and **3**. As a result, the composite chamber **26A** including the first and second chambers **24A** and **24B** and the two slits **25Aa** and **25Ab** is formed. The two slits **25Aa** and **25Ab** are arranged in the width direction that is orthogonal to the arrangement direction of the first and second chambers **24A** and **24B** and to the thickness direction of the back member **22** (body **20**). Further, the two slits **25Aa** and **25Ab** are located at both ends of the first and second chambers **24A** and **24B** in the width direction.

(25) As shown in FIG. **2**, a mode in which sixth and seventh chambers **24F** and **24G** are connected by a slit **25F** to form a composite chamber **26F**, a mode in which 12th and 13th chambers **24L** and **24M** are connected by a slit **25L** to form a composite chamber **26L**, a mode in which 15th and 16th chambers **24O** and **24P** are connected by a slit **25O** to form a composite chamber **26O**, and a mode in which 18th and 19th chambers **24R** and **24S** are connected by a slit **25R** to form a composite chamber **26R** are all the same as the mode in which the first and second chambers **24A** and **24B** are connected by the slits **25Aa** and **25Ab** to form the composite chamber **26A**.

(26) The third and fourth chambers **24C** and **24D** are connected by one slit **25C**. As a result, the composite chamber **26C** including the third and fourth chambers **24C** and **24D** and the one slit **25C** is formed. The one slit **25C** is located in the middle of the third and fourth chambers **24C** and **24D** in the width direction orthogonal to the arrangement direction of the third and fourth chambers **24C** and **24D** and to the thickness direction of the back member **22** (body **20**). The slit **25C** may be located at an end portion of the third and fourth chambers **24C** and **24D** in the width direction, for example.

(27) The 10th chamber **24J** is connected to the 9th chamber **24I** and the 11th chamber **24K** located on both sides thereof by slits **25I** and **25J**, respectively. That is, the 9th to 11th three chambers **24I** to **24K** are connected by the slits **25I** and **25J**. As a result, a composite chamber **26I** including the 9th to 11th chambers **24I** to **24K** and the slits **25I** and **25J** is formed. The mode in which the 9th and 10th chambers **24I** and **24J** are connected by the slits **25I** and the mode in which the 10th and 11th chambers **24J** and **24K** are connected by the slits **25J** are each the same as the mode in which the first and second chambers **24A** and **24B** are connected by the slits **25Aa** and **25Ab**.

(28) The fifth, eighth, 14th and 17th chambers **24E**, **24H**, **24N**, **24Q** are not connected to other chambers **24**.

(29) As described above, according to the body structure **2** of the present embodiment and the electric guitar **1** including the body structure **2**, by connecting the two chambers **24** with the slit **25**, the volume of the composite chamber **26** which includes the two chambers **24** and the slit **25** is

larger than the volume of each of the two chambers 24. As a result, the resonance frequency of the composite chamber 26 is lower than the resonance frequency of each of the two chambers 24. That is, by controlling the volumes of the chambers 24 and 26, it is possible to make the resonance frequencies of the plurality of chambers 24 and 26 formed in the body 20 different from each other. For example, although the resonance frequencies of two chambers 24 having substantially the same volume are approximately the same, by connecting these two chambers 24 with the slit 25, the number of chambers 24 having substantially the same resonance frequency can be reduced.

(30) This makes it possible to control the acoustic phenomenon of the body 20. Accordingly, even the body 20 of the electric guitar 1 having a plurality of weight-reducing chambers 24 can generate a beautiful and reverberant sound.

(31) Further, in the body structure 2 of the present embodiment, the cross-sectional area of the slit 25 connecting the two chambers 24 is smaller than the cross-sectional area of each of the two chambers 24. As a result, it is possible to secure a chamber having a large volume (that is, the composite chamber 26) while suppressing a decrease in the rigidity of the body 20.

(32) In the body structure 2 of the present embodiment, the resonance frequency of the composite chamber 26 including the two chambers 24 and the slit 25 can be controlled by appropriately changing the number of the slits 25 connecting the two chambers 24. Thereby it is possible to control the acoustic phenomenon of the body 20.

(33) Although the present disclosure has been described in detail above, the present disclosure is not limited to the above embodiments, and various modifications can be made without departing from the spirit of the present disclosure.

(34) In some embodiments of the present disclosure, the body structure 2 may have a sound absorbing material 27 housed in the slit 25, as shown for example in FIG. 4. The sound absorbing material 27 is a member that absorbs sound, such as urethane foam. In such a configuration, the resonance frequency of the composite chamber 26 including the two chambers 24 and the slit 25 connecting them can be controlled by the sound absorbing material 27. Thereby, the acoustic phenomenon of the body 20 can be controlled.

(35) In some embodiments of the present disclosure, the cross-sectional area of the slit 25 may be the same as, for example, the cross-sectional area of each of the two chambers 24.

(36) According to some embodiments of the present disclosure, it is possible to control the acoustic phenomenon of the body of an electric guitar having a plurality of chambers.

(37) The disclosure invention can be applied to an electric guitar, particularly the body of an electric guitar. According to the present invention, it is possible to control the acoustic phenomenon of the body of an electric guitar having a plurality of chambers.

Claims

1. A body structure of an electric guitar, comprising: a body comprising a top member and a back member that overlap to form the body, the back member comprising: a first chamber; a second chamber spaced apart from the first chamber; and a slit that connects the first chamber and the second chamber to each other, wherein the first chamber, the second chamber, and the slit do not open to outside of the body in a state where the top member overlaps with the back member to form the body, and wherein the first chamber and the second chamber are formed in a region of the body other than regions to which a neck, a bridge, electromagnetic pickups, and controllers are attached, when viewed from a thickness direction of the body.

2. The body structure of the electric guitar according to claim 1, wherein a cross-sectional area of the slit taken along a direction orthogonal to an arrangement direction of the first chamber and the second chamber is smaller than a cross-sectional area of the first chamber and a cross-sectional area of the second chamber taken along the direction orthogonal to the arrangement direction.

3. The body structure of the electric guitar according to claim 1, wherein a ratio of a volume of the

second chamber to a volume of the first chamber is 70% or more and 130% or less.

4. The body structure of the electric guitar according to claim 1, wherein the body comprises a plurality of slits connecting the first chamber and the second chamber to each other.
 5. The body structure of the electric guitar according to claim 1, further comprising a sound absorbing material disposed in the slit.
 6. An electric guitar comprising the body structure according to claim 1.
 7. The body structure of the electric guitar according to claim 1, wherein the top member has a first thickness and the back member has a second thickness greater than the first thickness.
 8. The body structure of the electric guitar according to claim 7, wherein the first chamber is a first recessed portion of a front surface of the back member that faces the top member, and the second chamber is a second recessed portion of the front surface of the back member that faces the top member.
 9. The body structure of the electric guitar according to claim 8, wherein the slit that connects the first chamber and the second chamber to each other is a third recessed portion of the front surface of the back member that faces the top member.
 10. The body structure of the electric guitar according to claim 9, wherein a depth of the third recessed portion equals a depth of the first recessed portion and a depth of the second recessed portion.
 11. The body structure of the electric guitar according to claim 9, wherein a depth of the third recessed portion is less than a depth of the first recessed portion and a depth of the second recessed portion.
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